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Trial Values

1st Pass Thru:

L S/N	$A_{11}+5$	$f_{11} = \frac{1}{2}$		
M S/N	$A_{11}+0$	A_{11}	1.0	
S.N Temp				
L S/N	$A_{11}+4$	$f_9 = \frac{1}{5}$	0.2	
M S/N	$A_{11}+1$	$f_{10} = \frac{1}{2}$	0.5	
A S/N (Temp)				
d S/N	$A_{11}+10$	$A_{11} =$		

Exact Value of one Term:

$$B_{44} = \begin{vmatrix} 1 & \frac{1}{5} & \frac{1}{9} \\ \frac{1}{2} & \frac{1}{6} & \frac{1}{10} \\ \frac{1}{3} & \frac{1}{7} & \frac{1}{11} \end{vmatrix}$$

$$\frac{1}{11}\left(1\cdot\frac{1}{6}-\frac{1}{2}\cdot\frac{1}{5}\right)-\frac{1}{10}\left(1\cdot\frac{1}{7}-\frac{1}{5}\cdot\frac{1}{3}\right)+\frac{1}{9}\left(\frac{1}{2}\cdot\frac{1}{7}-\frac{1}{3}\cdot\frac{1}{6}\right)$$

$$\frac{1}{11}\left(\frac{1}{6}-\frac{1}{10}\right)-\frac{1}{10}\left(\frac{1}{7}-\frac{1}{15}\right)+\frac{1}{9}\left(\frac{1}{14}-\frac{1}{18}\right)$$

$$\frac{1}{11}\left(\frac{4}{60}\right)-\frac{1}{10}\left(\frac{8}{105}\right)+\frac{1}{9}\left(\frac{4}{252}\right)$$

$$\frac{1}{11}\left(\frac{1}{15}\right)-\frac{1}{10}\left(\frac{8}{105}\right)+\frac{1}{9}\left(\frac{1}{63}\right)$$

$$\frac{1}{165}-\frac{4}{525}+\frac{1}{567}$$

$$\frac{1}{11\cdot5\cdot3}-\frac{4}{5\cdot7\cdot5\cdot3}+\frac{1}{3\cdot3\cdot7\cdot3\cdot3}$$

$$\frac{1}{3}\left[\frac{1}{55}-\frac{4}{175}+\frac{1}{189}\right]$$

$$\frac{1}{3}\left[\frac{(175)(189)-4(55)(189)+(55)(175)}{(55)(175)(189)}\right]$$

$$\frac{1}{3}\left[\frac{33075-41580+9625}{1819125}\right]$$

$$\frac{1}{3}\left[\frac{1120.}{1,819,125.}\right] = \frac{1120.}{5457375} = .2052267 \times 10^{-3}$$

$$= \frac{224.}{1,091,475} \overset{2\cdot2\cdot8\cdot7}{=} \frac{32}{155,925.}$$

Matrix

$$\begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{5} & \frac{1}{6} & \frac{1}{7} & \frac{1}{8} \\ \frac{1}{9} & \frac{1}{10} & \frac{1}{11} & \frac{1}{12} \\ \frac{1}{13} & \frac{1}{14} & \frac{1}{15} & \frac{1}{16} \end{vmatrix}$$

Transposed

$$\begin{vmatrix} 1 & \frac{1}{5} & \frac{1}{9} & \frac{1}{13} \\ \frac{1}{2} & \frac{1}{6} & \frac{1}{10} & \frac{1}{14} \\ \frac{1}{3} & \frac{1}{7} & \frac{1}{11} & \frac{1}{15} \\ \frac{1}{4} & \frac{1}{8} & \frac{1}{12} & \frac{1}{16} \end{vmatrix}$$

4 → I 2 → r 1 → r'

4 → J 2 → c 1 → c'

3 4 →

3 4

4(3-1) + 3-1 + A11 = A11+15

4(4-1) + 4-1 + B11 = B11+15

1 3-4

3-3

4-4

3-3

4(2-1) + 2-1 + A1 = A1+5

4(2-1) + 1-1 A1+4

4 + 1 A1+1

0 + 1-1 A1

$$B_{32} = -\begin{vmatrix} 1 & \frac{1}{5} & \frac{1}{13} \\ \frac{1}{3} & \frac{1}{7} & \frac{1}{15} \\ \frac{1}{4} & \frac{1}{8} & \frac{1}{16} \end{vmatrix} \qquad B_{23} = -\begin{vmatrix} 1 & \frac{1}{9} & \frac{1}{13} \\ \frac{1}{2} & \frac{1}{10} & \frac{1}{14} \\ \frac{1}{4} & \frac{1}{12} & \frac{1}{16} \end{vmatrix}$$

$$B_{32} = -\left[\frac{1}{16}\left(1\cdot\frac{1}{7} - \frac{1}{3}\cdot\frac{1}{5}\right) - \frac{1}{15}\left(1\cdot\frac{1}{8} - \frac{1}{4}\cdot\frac{1}{5}\right) + \frac{1}{13}\left(\frac{1}{3}\cdot\frac{1}{8} - \frac{1}{4}\cdot\frac{1}{7}\right)\right]$$

$$= \frac{1}{16}\left(\frac{1}{7} - \frac{1}{15}\right) - \frac{1}{15}\left(\frac{1}{8} - \frac{1}{20}\right) + \frac{1}{13}\left(\frac{1}{24} - \frac{1}{28}\right)$$

$\frac{1}{4}\left(\frac{1}{2} - \frac{1}{5}\right)$ $\qquad$ $\frac{1}{4}\left(\frac{1}{6} - \frac{1}{7}\right)$

$$= \frac{1}{2}\left(\frac{1}{105}\right) - \frac{1}{20}\left(\frac{1}{10}\right) + \frac{1}{52}\left(\frac{1}{42}\right)$$

$$= \frac{1}{210} - \frac{1}{200} + \frac{1}{2184}$$

$$= -\left[\frac{-1}{4200} + \frac{1}{2184}\right]$$

$$= \frac{-2016}{9,172,800}$$

$$B_{32} = -.2197802197 \times 10^{-3}$$

$$B_{23} = -\left[\frac{1}{16}\left(1\cdot\frac{1}{10} - \frac{1}{2}\cdot\frac{1}{9}\right) - \frac{1}{14}\left(1\cdot\frac{1}{12} - \frac{1}{4}\cdot\frac{1}{9}\right) + \frac{1}{13}\left(\frac{1}{2}\cdot\frac{1}{12} - \frac{1}{4}\cdot\frac{1}{10}\right)\right]$$

$\frac{1}{2}\cdot\frac{1}{5}$ $\qquad$ $\frac{1}{3\cdot4}$ $\qquad$ $\frac{1}{4\cdot3\cdot3}$ $\qquad$ $\frac{1}{2\cdot3\cdot4}$ $\qquad$ $\frac{1}{4\cdot2\cdot5}$

$$= \frac{1}{32}\left(\frac{1}{5} - \frac{1}{9}\right) - \frac{1}{168}\left(1 - \frac{1}{3}\right) + \frac{1}{104}\left(\frac{1}{3} - \frac{1}{5}\right)$$

$$= \frac{1}{32}\left(\frac{4}{45}\right) - \frac{1}{168}\left(\frac{2}{3}\right) + \frac{1}{104}\left(\frac{2}{15}\right)$$

$$= \frac{1}{8}\cdot\frac{1}{45} - \frac{1}{84}\cdot\frac{1}{3} + \frac{1}{52}\cdot\frac{1}{15}$$

$$= \frac{1}{360} - \frac{1}{252} + \frac{1}{780} \qquad = -\frac{108}{90720} + \frac{1}{780} = \frac{-27}{22,680} + \frac{1}{780}$$

$$= -\left\{\frac{-1}{840} + \frac{1}{780}\right\} = -\left[\frac{60}{655,200}\right]$$

$$B_{23} = -.915750915 \times 10^{-4}$$

$$B_{43} = -\begin{vmatrix} 1 & \frac{1}{5} & \frac{1}{9} \\ \frac{1}{2} & \frac{1}{6} & \frac{1}{10} \\ \frac{1}{4} & \frac{1}{8} & \frac{1}{12} \end{vmatrix} \qquad B_{42} = \begin{vmatrix} 1 & \frac{1}{5} & \frac{1}{9} \\ \frac{1}{3} & \frac{1}{7} & \frac{1}{11} \\ \frac{1}{4} & \frac{1}{8} & \frac{1}{12} \end{vmatrix}$$

$$= \frac{1}{12}\left(1\cdot\frac{1}{6} - \frac{1}{2}\cdot\frac{1}{5}\right) - \frac{1}{10}\left(1\cdot\frac{1}{8} - \frac{1}{4}\cdot\frac{1}{5}\right) + \frac{1}{9}\left(\frac{1}{2}\cdot\frac{1}{8} - \frac{1}{4}\cdot\frac{1}{6}\right)$$

$$= \frac{1}{24}\left(\frac{1}{3} - \frac{1}{5}\right) - \frac{1}{40}\left(\frac{1}{2} - \frac{1}{5}\right) + \frac{1}{72}\left(\frac{1}{2} - \frac{1}{3}\right)$$

$$= \frac{1}{24}\left(\frac{2}{15}\right) - \frac{1}{40}\left(\frac{3}{10}\right) + \frac{1}{72}\left(\frac{1}{6}\right) = .0055555555 - .007500 + .0023148148$$

$$= \frac{1}{180} - \frac{3}{400} + \frac{1}{432} = \frac{1}{20}\left[\frac{1}{9} - \frac{3}{20}\right] + \frac{1}{432}$$

$$= \frac{1}{20}\cdot\left(\frac{-7}{180}\right) + \frac{1}{20}\cdot\left(\frac{5}{108}\right) = \frac{1}{20}\left[\frac{756 - 900}{19,440}\right] = \frac{1}{20}\left[\frac{-144}{19,440}\right] = \frac{4\times X}{5(2160\times X)}$$

$$= \frac{1}{5(540)} = \frac{1}{2700}$$

$B_{43} = +\,.37037037 \times 10^{-3}$

$$B_{42} = \frac{1}{12}\left(1\cdot\frac{1}{7} - \frac{1}{3}\cdot\frac{1}{5}\right) - \frac{1}{11}\left(1\cdot\frac{1}{8} - \frac{1}{4}\cdot\frac{1}{5}\right) + \frac{1}{9}\left(\frac{1}{3}\cdot\frac{1}{8} - \frac{1}{4}\cdot\frac{1}{7}\right)$$

$$= \frac{1}{12}\left(\frac{1}{7} - \frac{1}{15}\right) - \frac{1}{44}\left(\frac{1}{2} - \frac{1}{5}\right) + \frac{1}{36}\left(\frac{1}{6} - \frac{1}{7}\right)$$

$$= \frac{1}{12}\left(\frac{8}{105}\right) - \frac{1}{44}\left(\frac{3}{10}\right) + \frac{1}{36}\left(\frac{1}{42}\right)$$

$$= \frac{2}{315} - \frac{3}{440} + \frac{1}{1512} = \frac{880 - 945}{138,600} + \frac{1}{1512} \quad (-65)$$

$$= \frac{-13}{27,720} + \frac{1}{1,512} = \frac{-1956 + 27,720}{41,912,640} = \frac{8,064}{41,912,640}$$

$B_{42} = 0,192400192 \times 10^{-3}$ $\qquad = \frac{1}{5197.5}$

$$B_{41} = -\begin{vmatrix} \frac{1}{2} & \frac{1}{6} & \frac{1}{10} \\ \frac{1}{3} & \frac{1}{7} & \frac{1}{11} \\ \frac{1}{4} & \frac{1}{8} & \frac{1}{12} \end{vmatrix}$$

$$= \frac{1}{12}\left(\frac{1}{2}\cdot\frac{1}{7} - \frac{1}{3}\cdot\frac{1}{6}\right) - \frac{1}{11}\left(\frac{1}{2}\cdot\frac{1}{8} - \frac{1}{4}\cdot\frac{1}{6}\right) + \frac{1}{10}\left(\frac{1}{3}\cdot\frac{1}{8} - \frac{1}{4}\cdot\frac{1}{7}\right)$$

$$= \frac{1}{24}\left(\frac{1}{7} - \frac{1}{9}\right) - \frac{1}{88}\left(\frac{1}{2} - \frac{1}{3}\right) + \frac{1}{40}\left(\frac{1}{6} - \frac{1}{7}\right)$$

$$= \frac{1}{12}\left(\frac{1}{63}\right) - \frac{1}{88}\left(\frac{1}{6}\right) + \frac{1}{40}\left(\frac{1}{42}\right) \qquad = .0013227513 - .0018939394 + .0005952381$$

$$= \frac{1}{12}\left[\frac{1}{63} - \frac{1}{44}\right] + \frac{1}{40}\left(\frac{1}{42}\right) = \frac{1}{12}\left[\frac{-19}{2772}\right] + \frac{1}{40}\left(\frac{1}{42}\right) \qquad = .0000245000$$

$$= \frac{1}{6\cdot4\cdot7}\left[\frac{-19}{198} + \frac{1}{10}\right]$$

$$= \frac{1}{168}\left[\frac{-190+198}{1980}\right] = \frac{1}{21}\left(\frac{1}{1980}\right) = \frac{1}{41{,}580}$$

$B_{41} = -.240500240 \times 10^{-4}$

$$|9| = \frac{1}{16}B_{44} + \frac{1}{15}B_{43} + \frac{1}{14}B_{42} + \frac{1}{13}B_{41}$$

$$= \left(\frac{2}{155{,}925}\right) + \left(\frac{-1}{40{,}500}\right) + \left(\frac{1}{72{,}765}\right) + \left(\frac{-1}{540{,}540}\right)$$

$= .0000128266794\#$
$-.0000246913580\,2$
$\quad .0000137428708\,8$
$-.0000001850001\,85$

$.0000 0002819050$

with 8 digit nos.

$$
\begin{aligned}
|A| &= (.625)(.20522687) && = .12826679 \\
&-(.66666667)(.37037037) && = -.24691358 \\
&+(.71428571)(.19240019) && = .13742871 \\
&=(.76923077)(.24050024) && = \underline{-.01850002} \\
& && \quad .28190
\end{aligned}
$$